

# Animal Classification With RPi Camera For Survey And Monitoring

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Hunting continues to make many animal species extinct, and the governments are unable to do much except introducing some laws and conducting regular surveys. But conducting a survey is not an easy task, especially without the help of technology.

The smart camera in this project uses Python code that has pre-trained TensorFlow models for forest monitoring and survey. The models help us in matching and mapping the image in frame with the model's trained data and thus in identifying those animals.

The images of elephants being detected with the help of author's prototype is shown in Fig. 1. The author's prototype that uses a Raspberry Pi (or RPi) with a camera module is shown in Fig. 2.

The Raspberry Pi captures a video image using the camera, then using the OpenCV module it cuts the video into frames. Next, using OpenCV and other modules, it processes the image of detected species and maps it with a category and pre-trained detection model. If the detected animal species is not there in the category list, it gives the output as 'none' and saves that image with data and timestamp in database folder so that it can be studied later.

If the detected animal species matches with any of the animals listed in *new.pbtxt* database, it identifies the name of the animal and saves it in text file *forestdb.txt* with time and date. It also saves the image of that animal with a time stamp in another folder for future reference.

In this way the smart camera keeps track and creates a database with pictures of animals present in a forest. It can also help in studying animal behaviour and the presence and growth of flora and fauna in the forest.

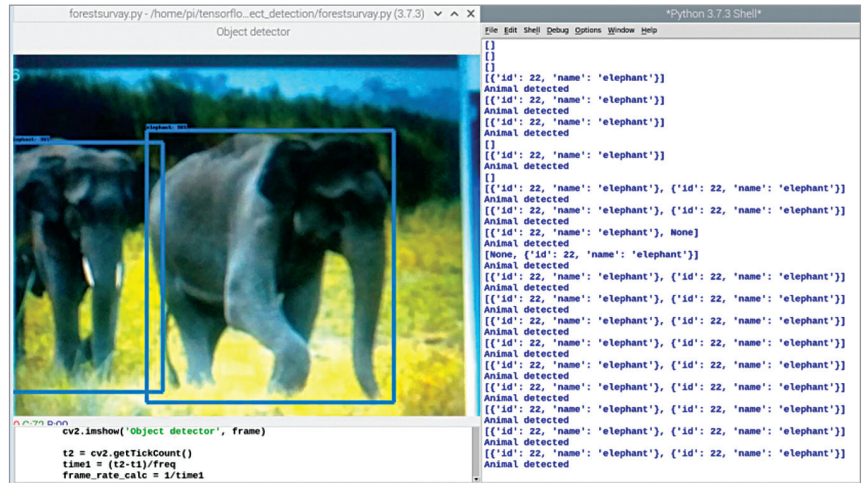


Fig. 1: Elephants being captured by author's prototype



Fig. 2: Author's prototype

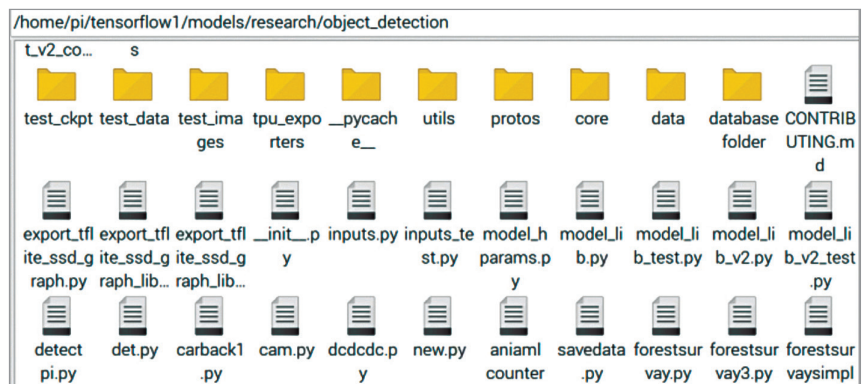


Fig. 3: Object\_detection folder